

CS 305: Computer Networks

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Lecture 2: Introduction

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Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

The network core

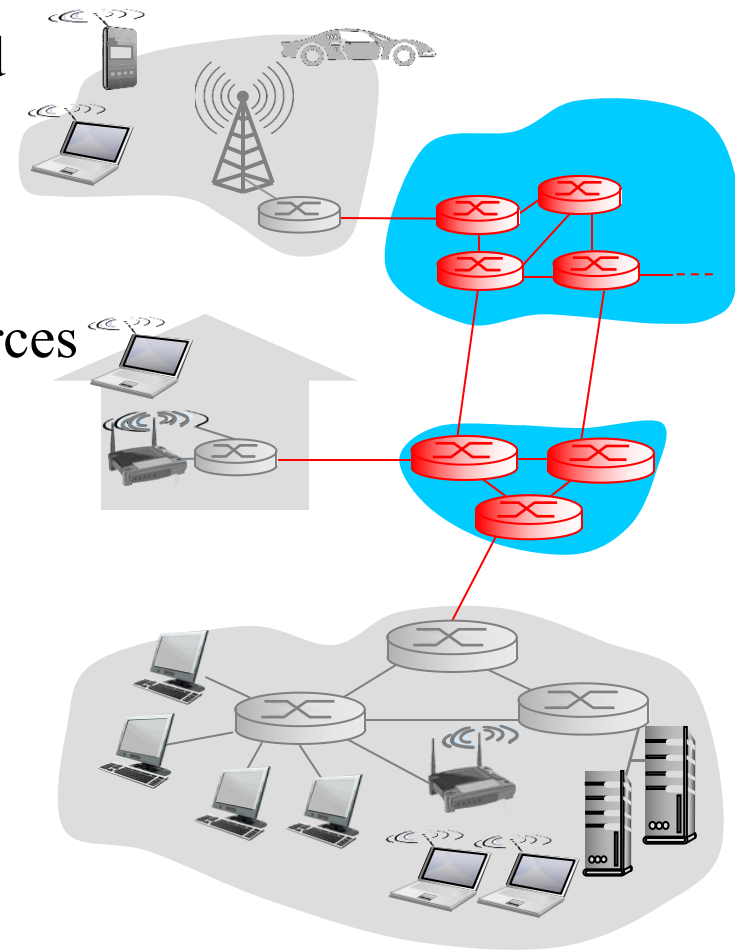
Network Core: The mesh of **packet switches** and **links** that interconnects the Internet's end systems.

How to exchange messages?

- ❖ Packet switching: not reserve (预留) resources
- ❖ Circuit switching (e.g., telephone): reserve

Packet switching: hosts break long messages into **packets**

- Each packet travels through communication links and packet switches
- Forward packets from one router to the next, across links on path from source to destination
- Each packet transmitted **at full link capacity**



Packet switching versus circuit switching

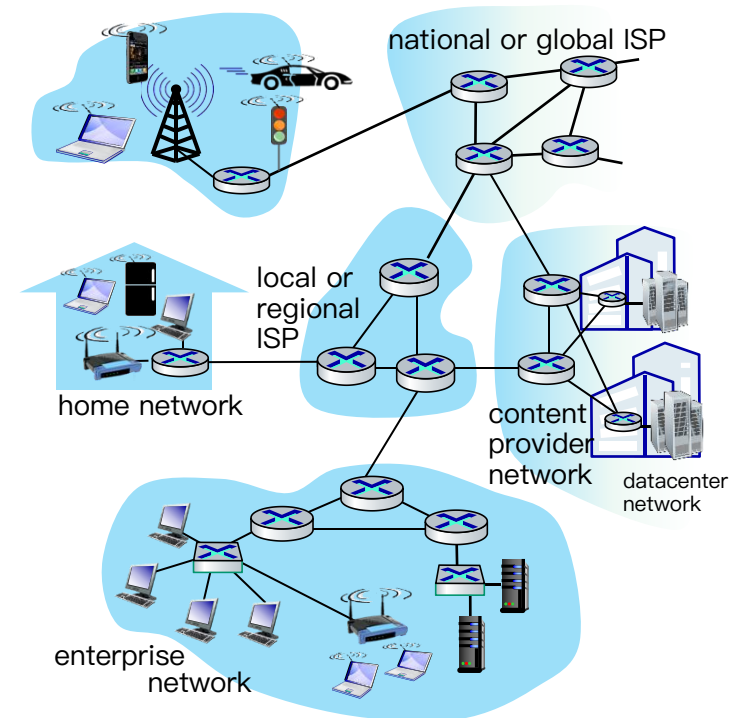
Packet switching:

- ❖ great for bursty data
 - resource sharing
 - simpler, no call setup
- ❖ **excessive congestion possible:** packet delay and loss
 - protocols needed for reliable data transfer, congestion control
- ❖ **Q: How to provide circuit-like behavior?**
 - bandwidth guarantees needed for audio/video apps
 - still an unsolved problem (chapter 7)

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

Internet structure: network of networks

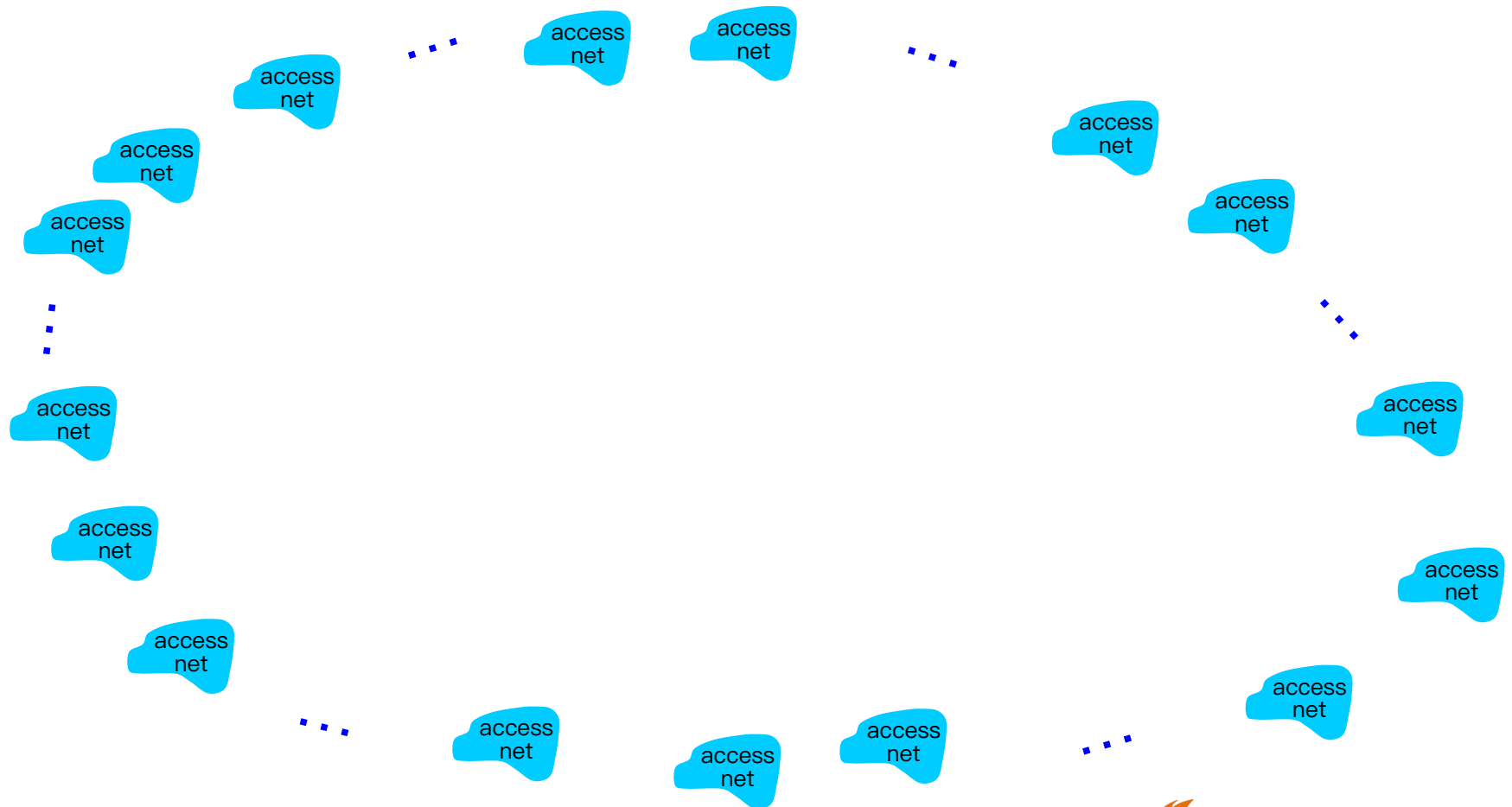
- ❖ End systems connect to Internet via **access ISPs** (Internet Service Providers)
 - Residential, company and university ISPs
 - E.g., China Mobile, university.
- ❖ Access ISPs in turn must be **interconnected**.
 - ❖ So that any two hosts can send packets to each other
- ❖ Resulting network of networks is very complex
 - ❖ Evolution was driven by **economics** and **national policies**



Let's take a stepwise approach to describe current Internet structure.

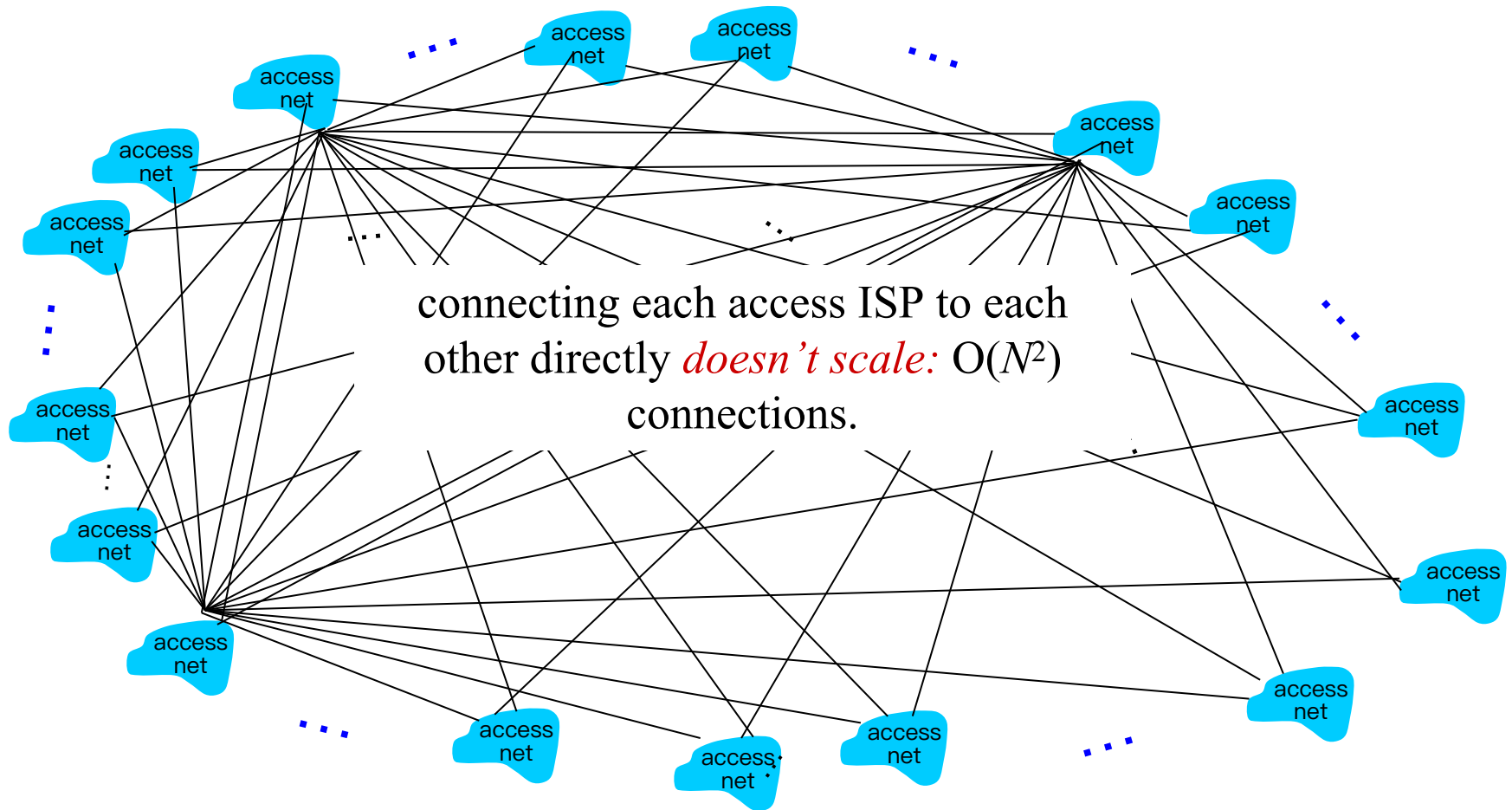
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



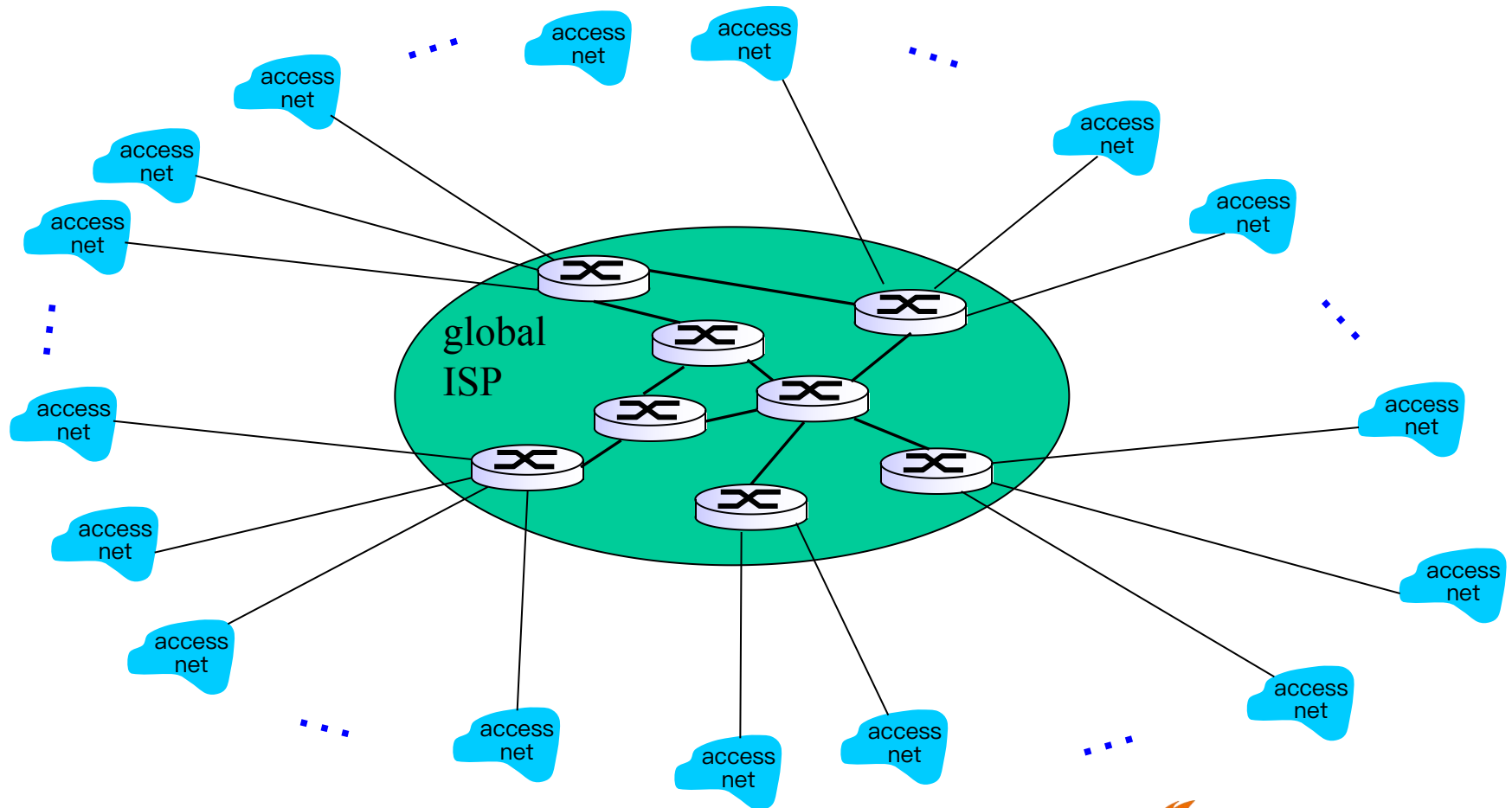
Internet structure: network of networks

Option 1: connect each access ISP to every other access ISP?



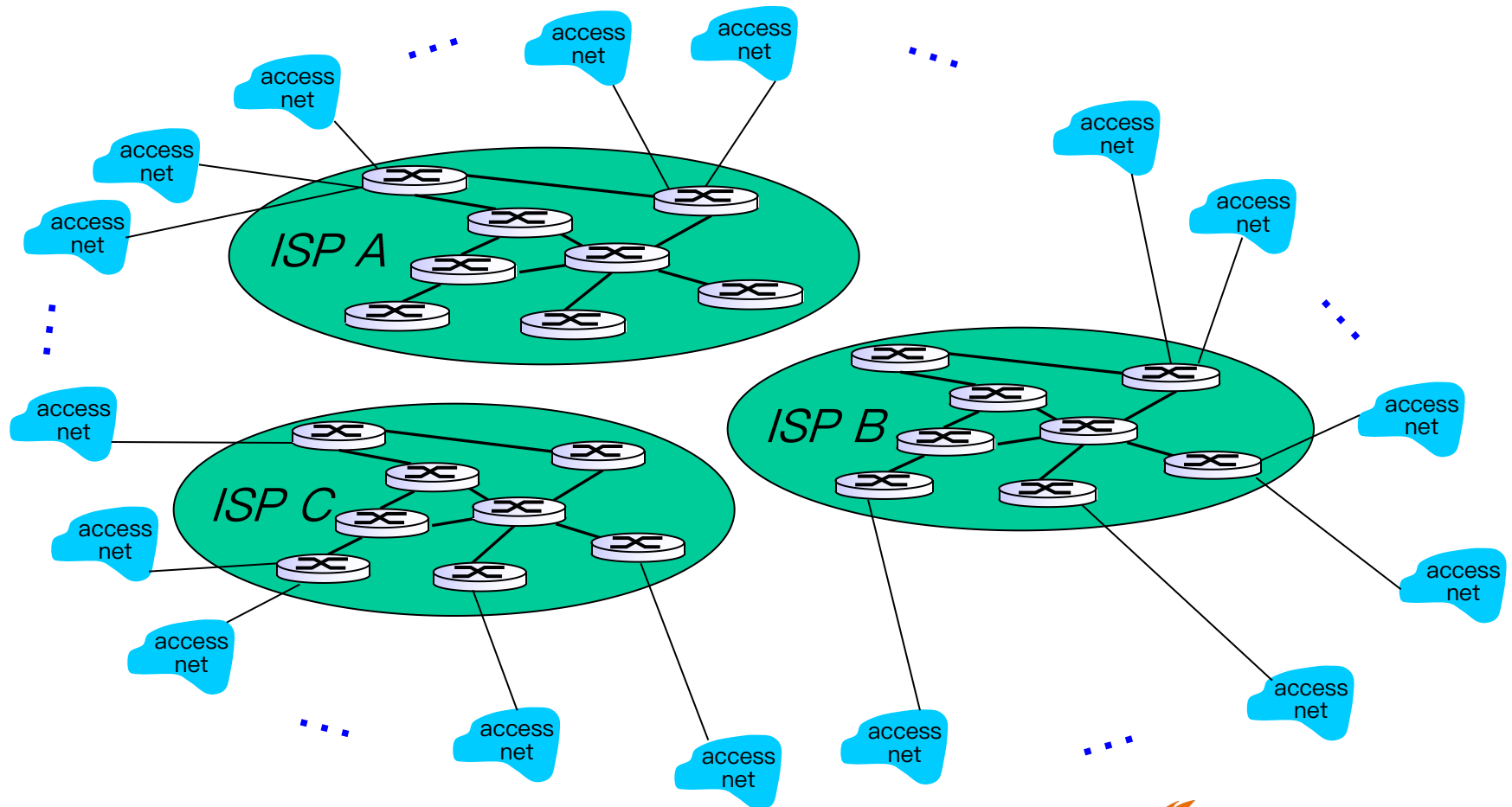
Internet structure: network of networks

Option 2: connect each access ISP to a global transit ISP? **Customer** and **provider** ISPs have economic agreement.



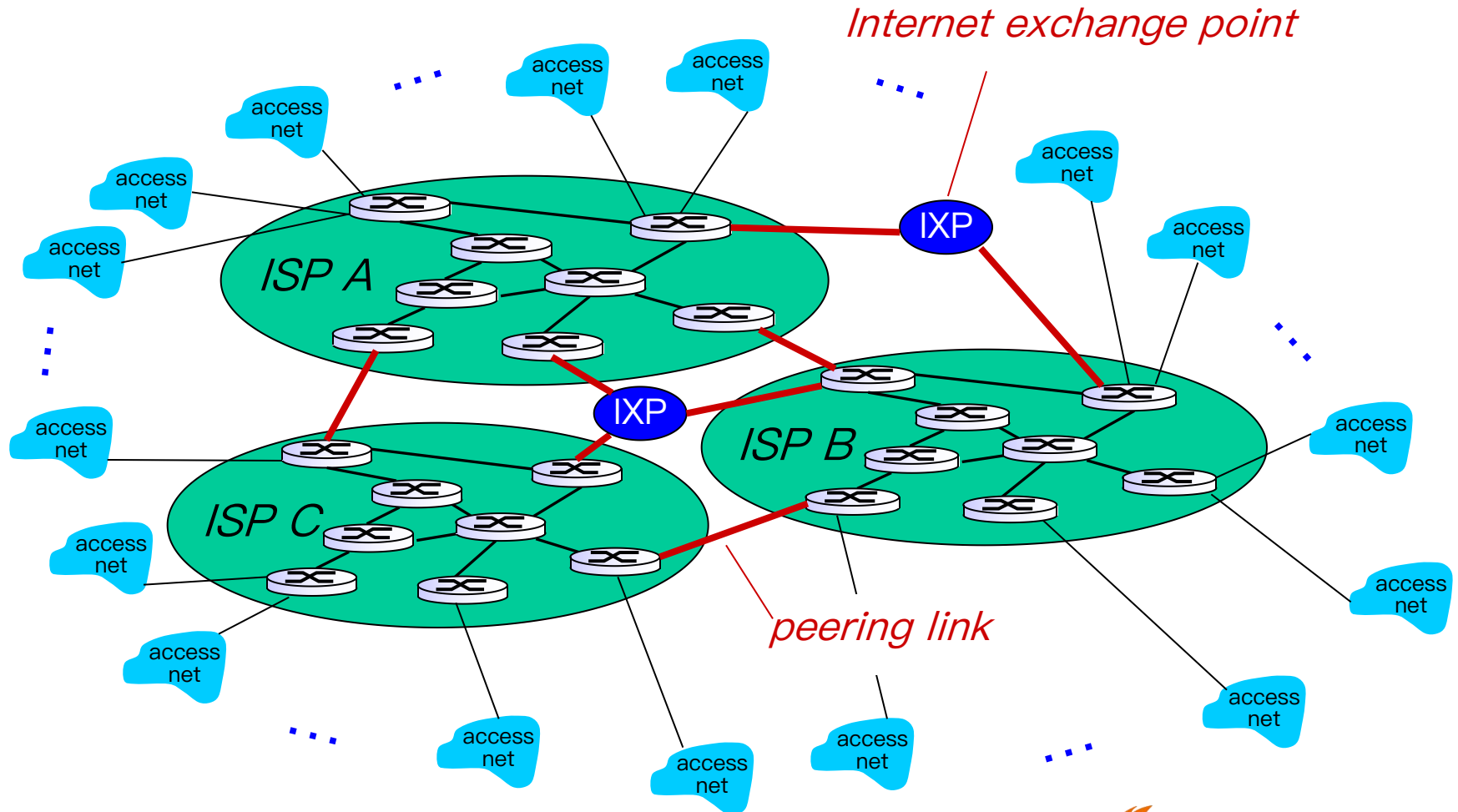
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors



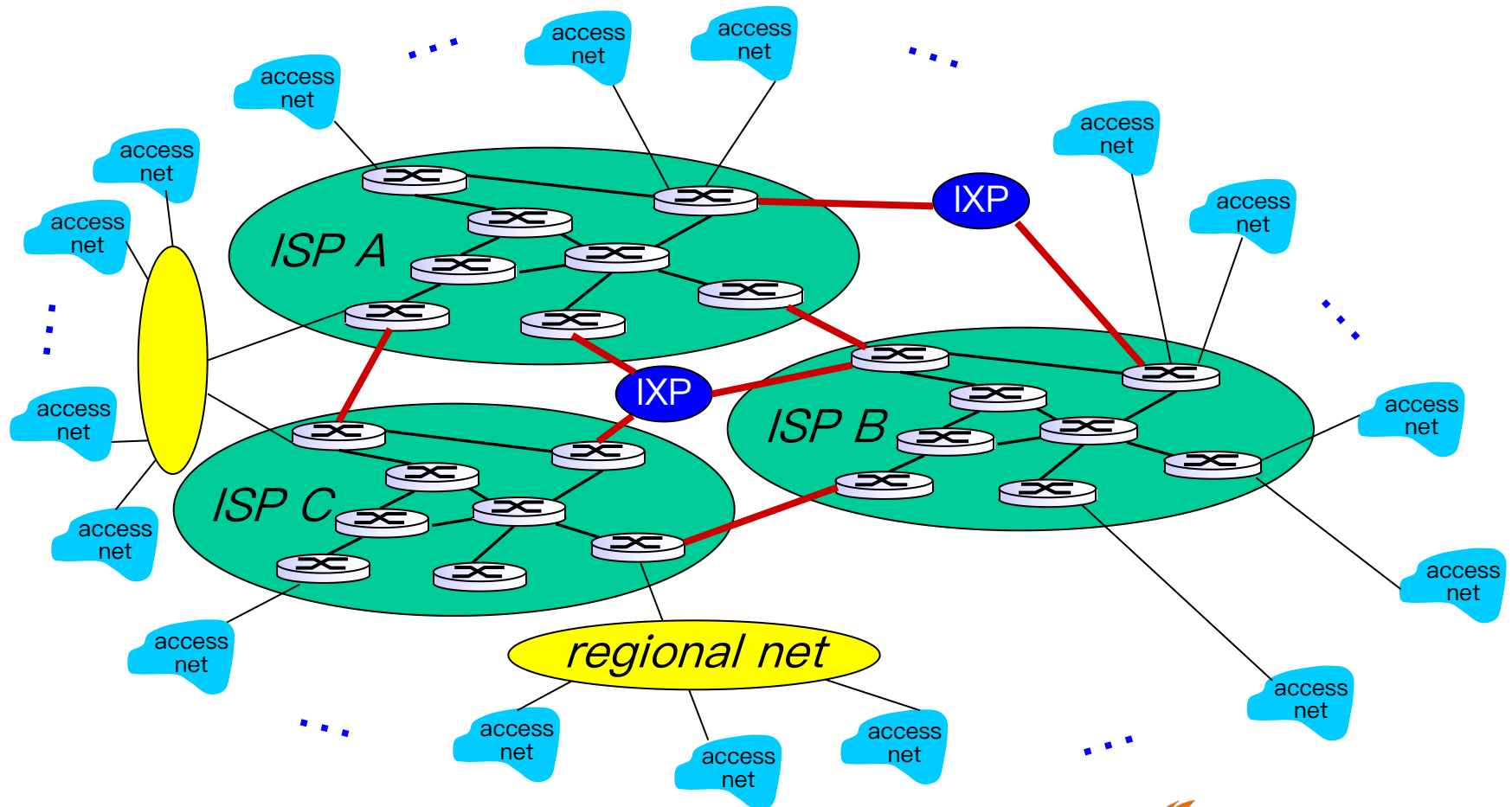
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors which must be interconnected



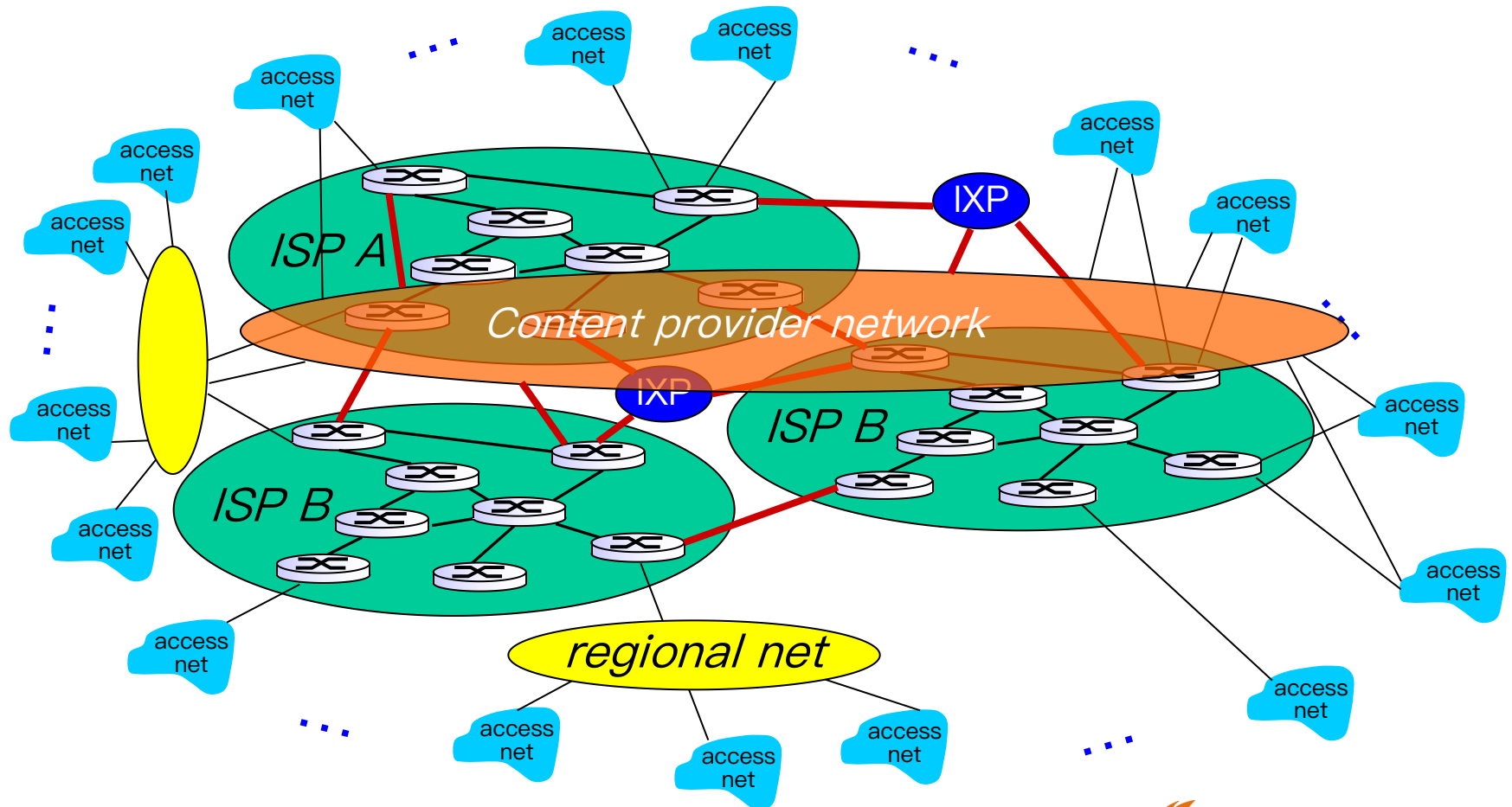
Internet structure: network of networks

... and regional networks may arise to connect access nets to ISPS

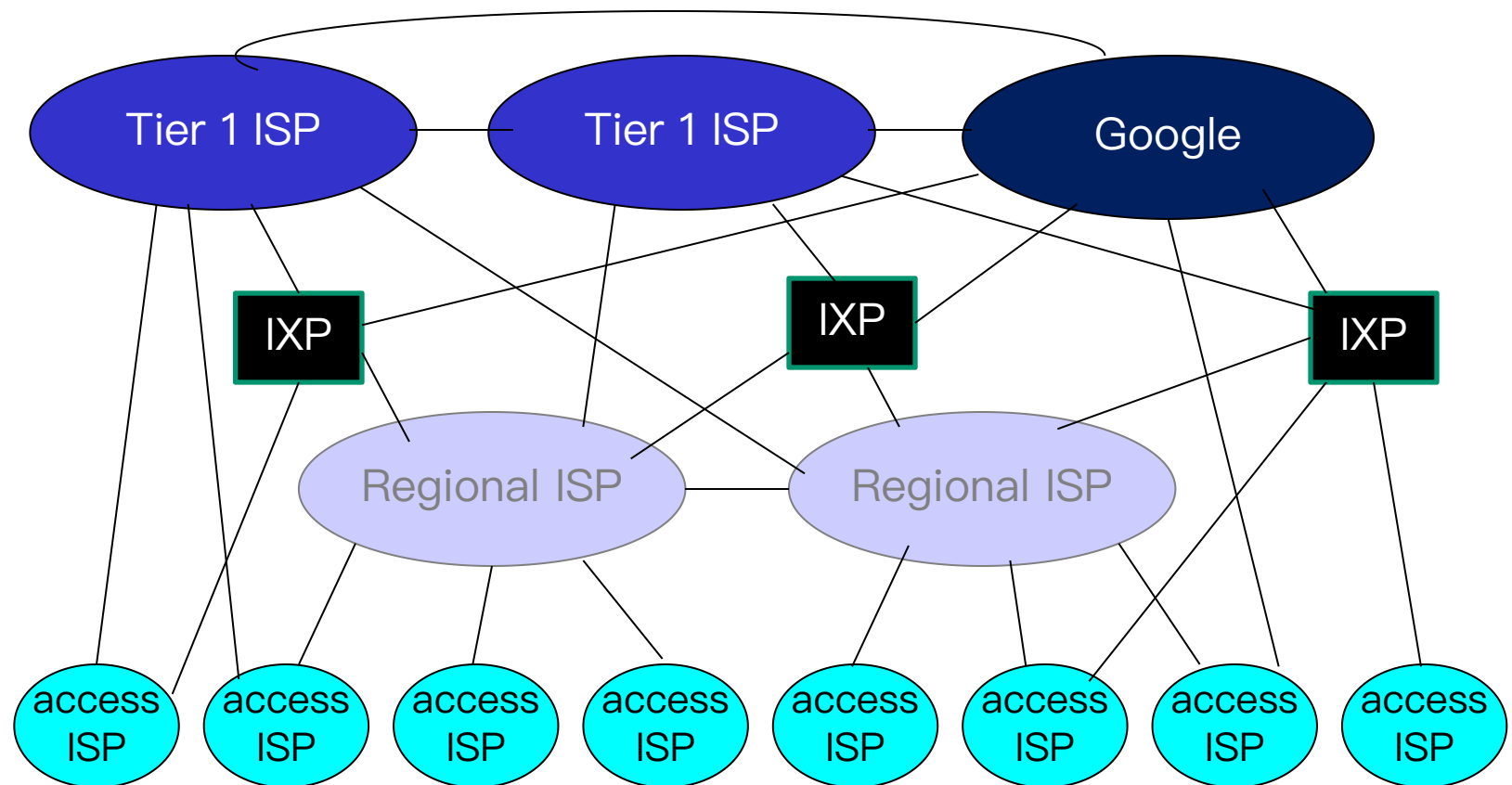


Internet structure: network of networks

... and content provider networks (e.g., Google, Microsoft) may run their own network, to bring services, content close to end users



Internet structure: network of networks



- “Tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- Multi-homing: to connect to two or more provider ISPs
- Peer: a pair of nearby ISPs at the same level
- Content provider network (e.g., Google): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

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1.7 history

Performance Metric

We would like Internet services to be able to **move as much data** as we want **between any two end systems**, instantaneously, **without any loss of data**.

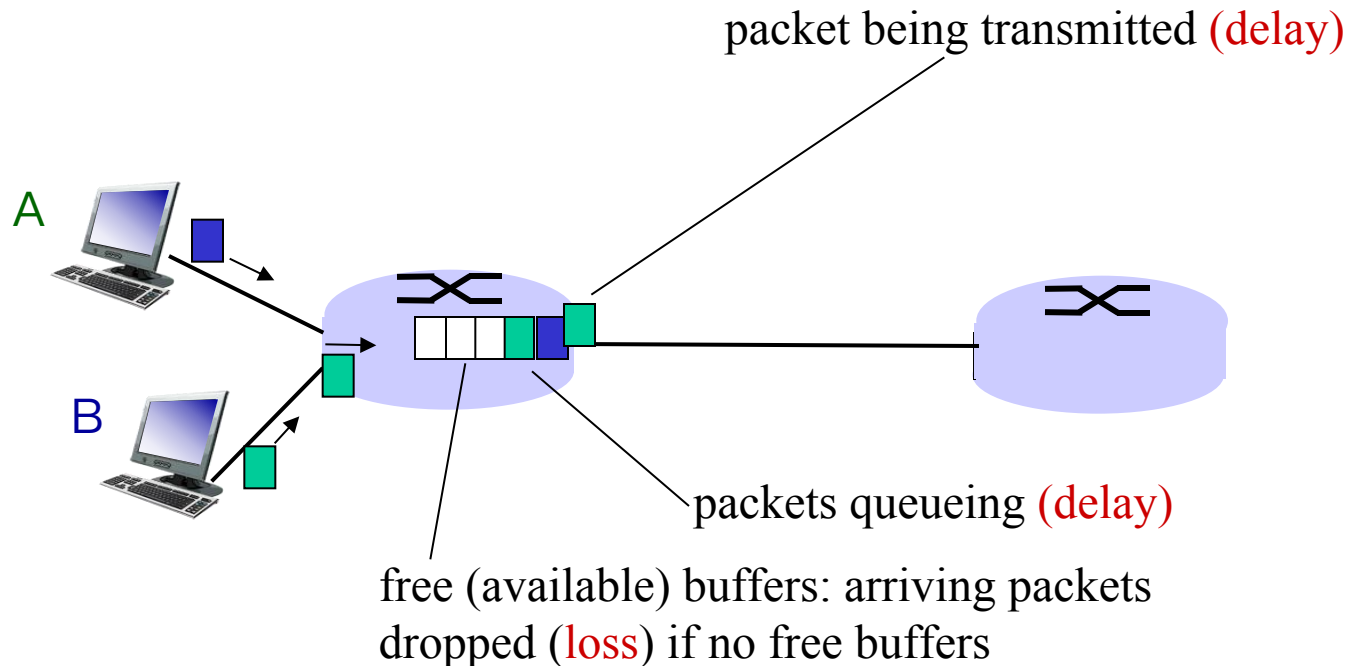
In Packet-Switched Networks:

- ❖ **Throughput**: the amount of data per second that can be transferred between end systems
- ❖ **Delay**
- ❖ **Packet loss**

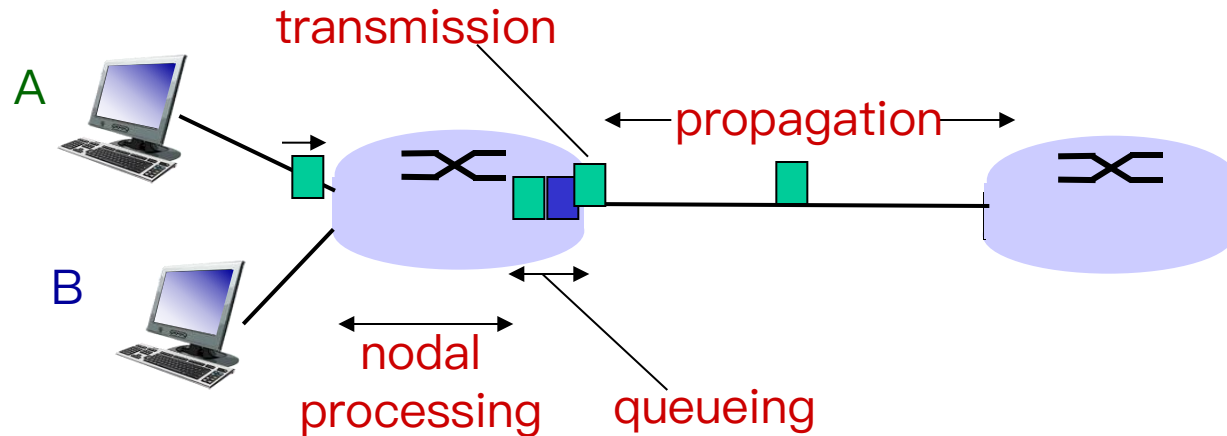
How do loss and delay occur?

Packets **queue** in router buffers

- ❖ packet arrival rate to link (temporarily) exceeds output link capacity
- ❖ packets queue, wait for turn



Four sources of packet delay



Nodal (节点) delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

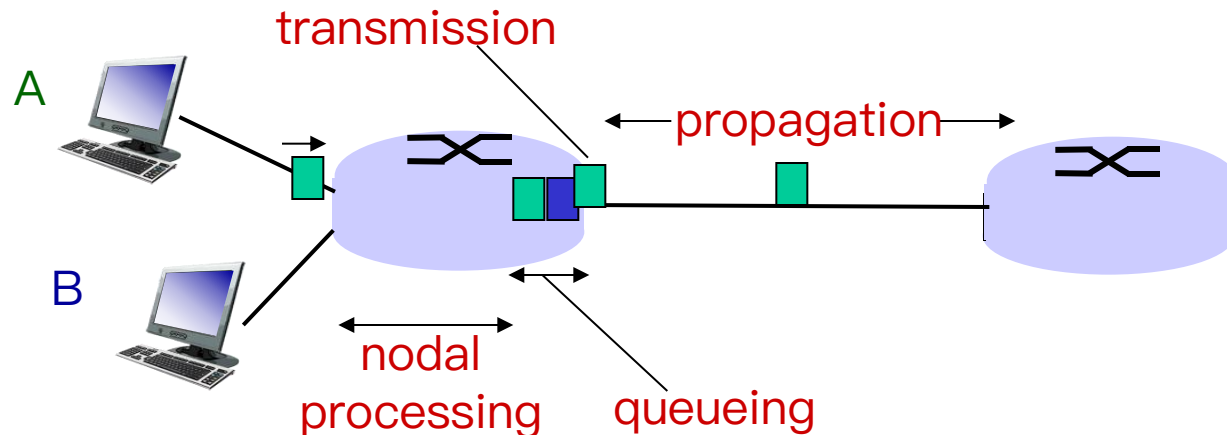
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of packet delay



Nodal (节点) delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay (传输时延):

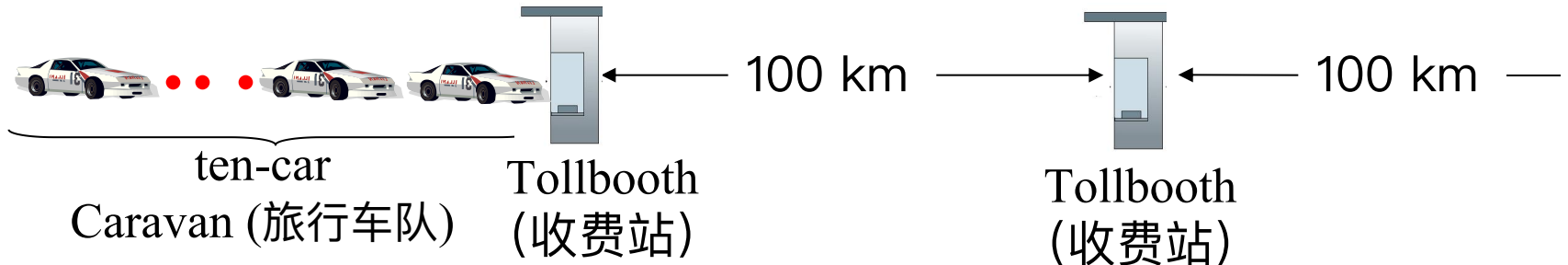
- L : packet length (bits)
- R : transmission rate (bps)
- $d_{\text{trans}} = L/R$

d_{prop} : propagation delay (传播时延):

- d : length of physical link
- s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

d_{trans} and d_{prop}
very different

Caravan analogy



- ❖ Cars “propagate” at 100 km/hr
- ❖ Tollbooth takes 12 sec to service one car (bit transmission time)
- ❖ car~bit; caravan ~ packet

Q: How long does it take the last car to arrive at the 2nd tollbooth?

- **Transmission delay:** time to “push” entire caravan through tollbooth onto highway = $12 \times 10 = 120$ sec
- **Propagation delay:** time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1$ hr
- *A:* 62 minutes

Queueing delay (revisited)

Nodal (节点) delay

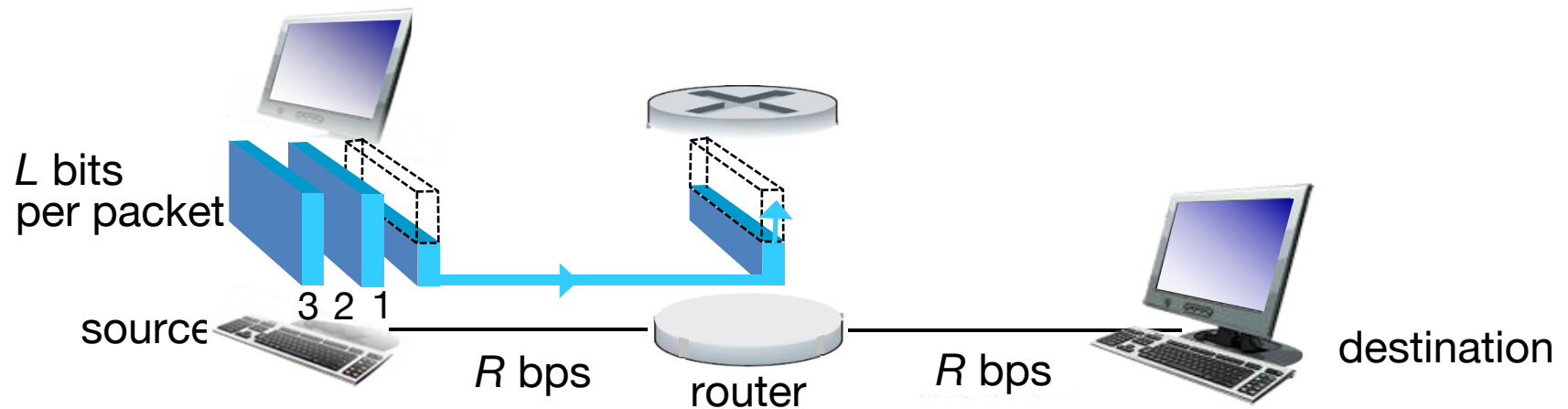
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Unlike the other three delays (namely, d_{proc} , d_{trans} , d_{prop}), the queuing delay can vary from packet to packet.

When characterizing queuing delay, statistical measures:

- average queuing delay
- variance of queuing delay
- the probability that the queuing delay exceeds some specified value

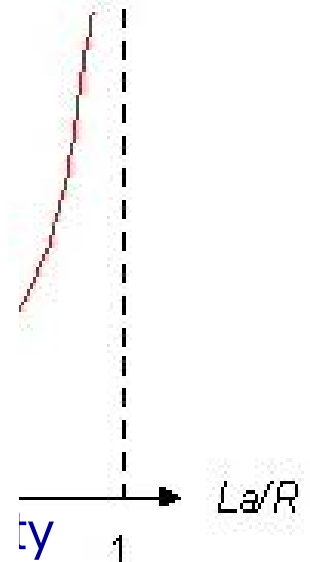
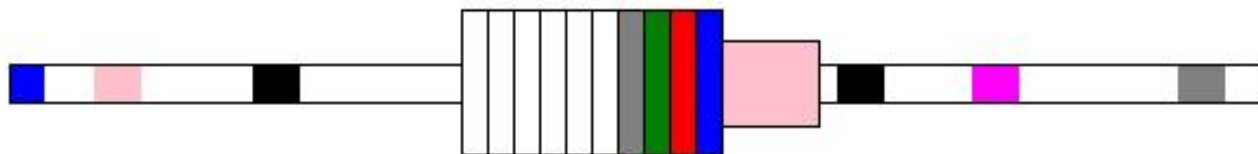
Queueing delay (revisited)



- ❖ R : transmission rate (bps)
- ❖ L : packet length (bits)
- ❖ a : average packet arrival rate (packets per sec)

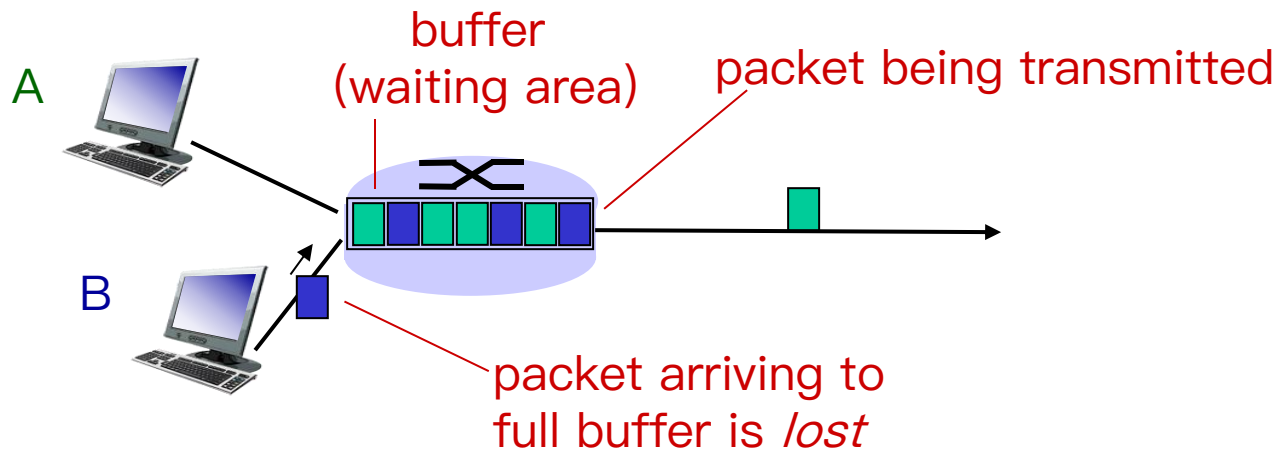
queueing
delay

Emission rate Transmission rate

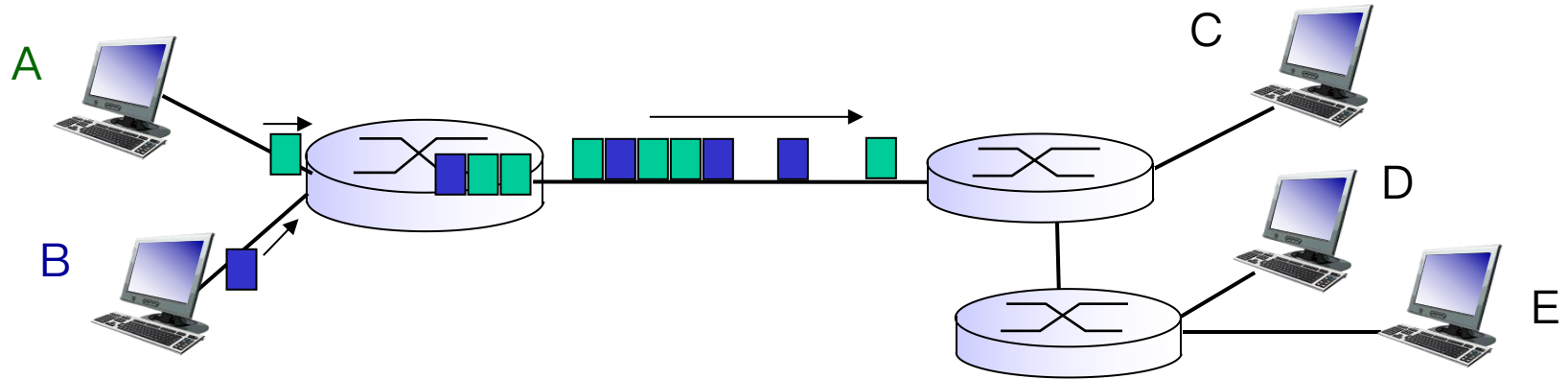


Packet loss

- ❖ Queue (aka buffer) preceding link in buffer has finite capacity
- ❖ Packet arriving to full queue will be **dropped** (aka lost)
- ❖ Lost packet may be retransmitted by previous node, by source end system, or not at all



End-to-End Delay



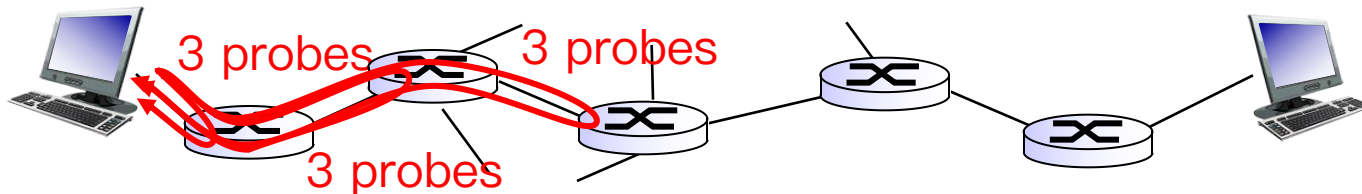
Suppose d_{queue} is negligible; $N-1$ routers between two hosts

End-to-End delay

$$d_{\text{end-end}} = N(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$$

“Real” Internet delays and routes

- ❖ What do “real” Internet delay & loss look like?
- ❖ **Traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply.



$N-1$ routers, send $3N$ packets

“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
```

trans-oceanic link

looks like delays
decrease! Why?

* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

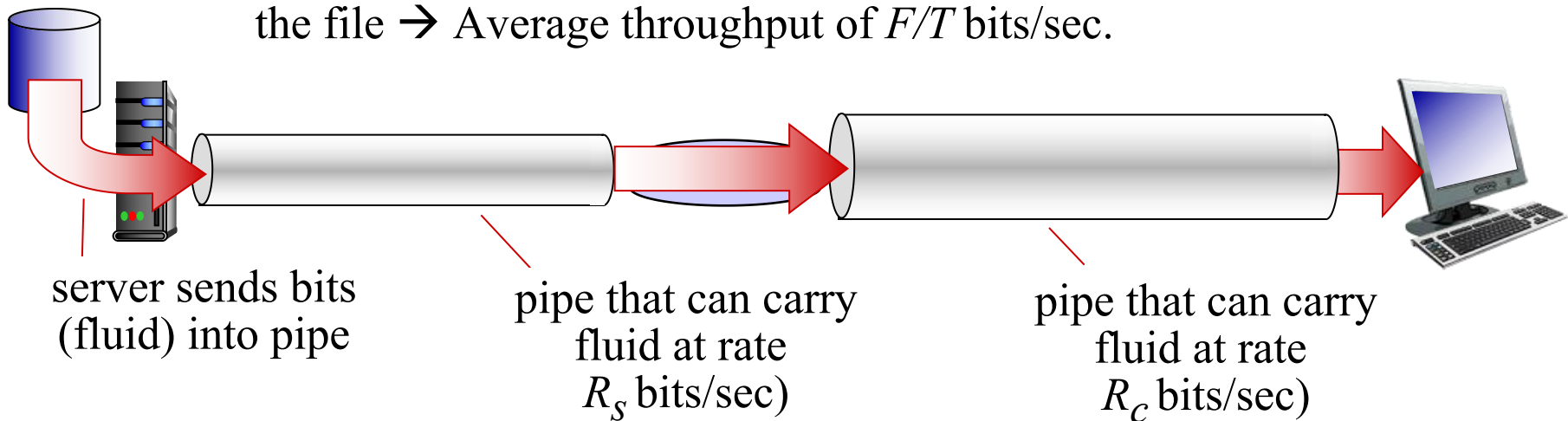
Throughput

❖ **Throughput:** rate (bits/time unit) at which bits transferred between sender/receiver

- *instantaneous:* rate at given point in time
- *average:* rate over longer period of time

The rate (bits/time unit) at which the receiver is receiving the file.

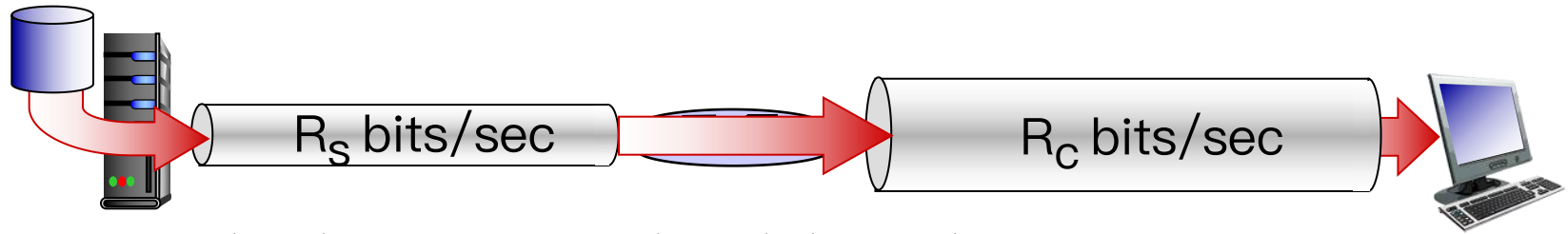
- A file of F bits; it takes T seconds for the receiver to receive the file $\rightarrow$ Average throughput of F/T bits/sec.



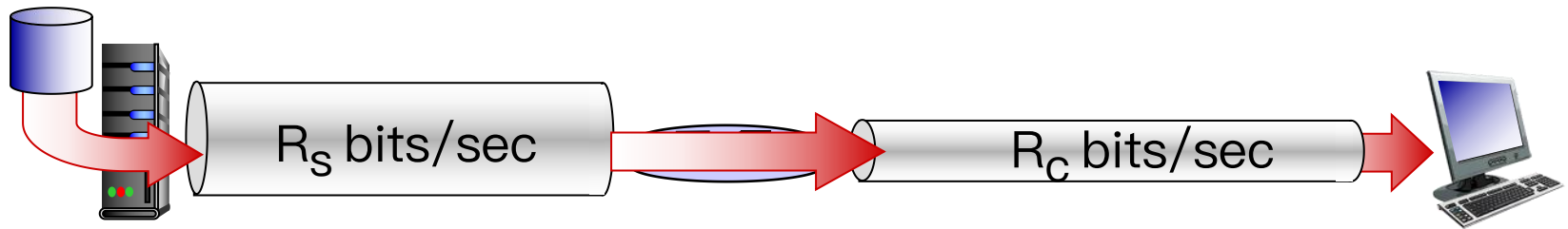
What is the server-to-client throughput?

Throughput (more)

❖ $R_s < R_c$ What is average end-end throughput?



❖ $R_s > R_c$ What is average end-end throughput?

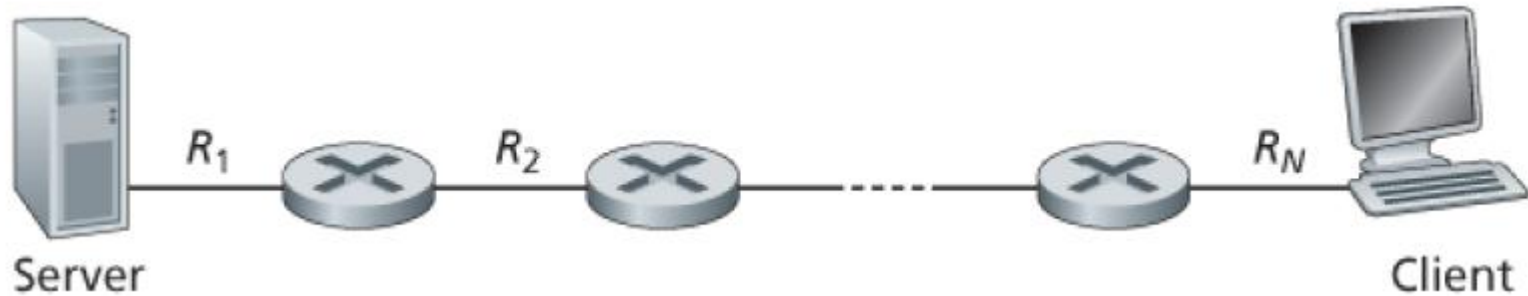


$\min\{R_s, R_c\}$

bottleneck link

link on end-end path that constrains end-end throughput

Throughput (more)



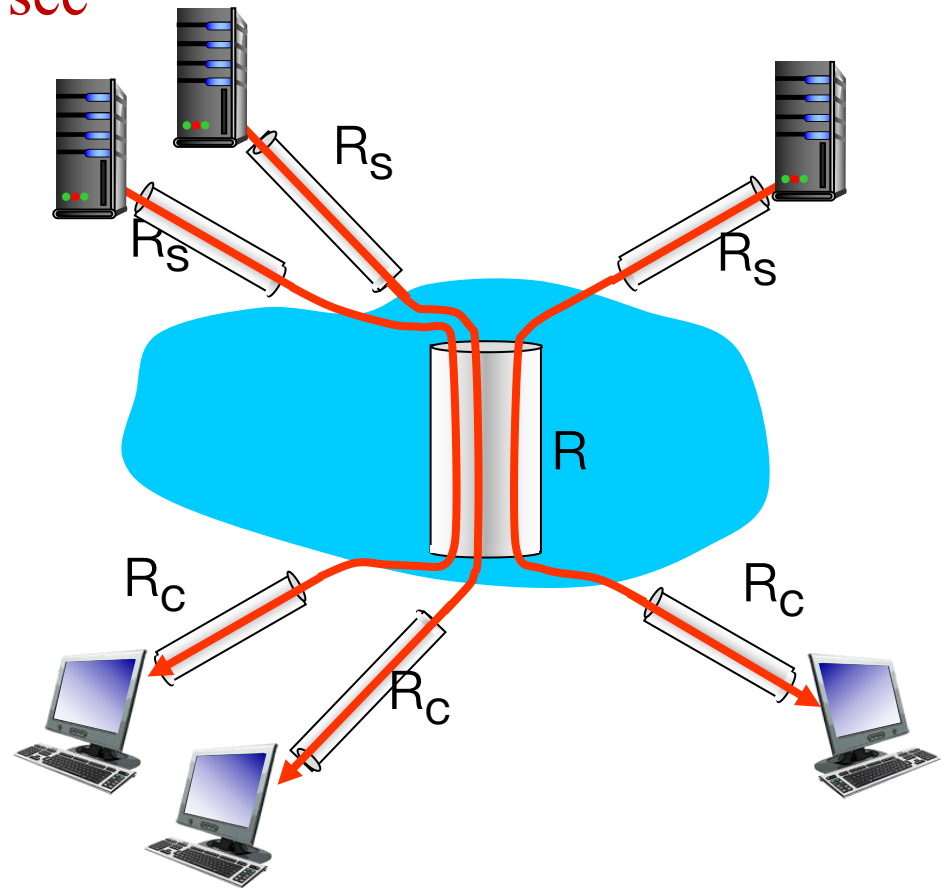
What is average end-end throughput?

$$\min\{R_1, R_2, \dots, R_N\}$$

Throughput: Internet scenario

10 connections (fairly) share
backbone bottleneck link R bits/sec

- ❖ Per-connection end-end throughput:
 $\min\{R_c, R_s, R/10\}$
- ❖ In practice: R_c or R_s is often bottleneck



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1.6 networks under attack: security

Protocol “layers”

Networks are complex,
with many “pieces” :

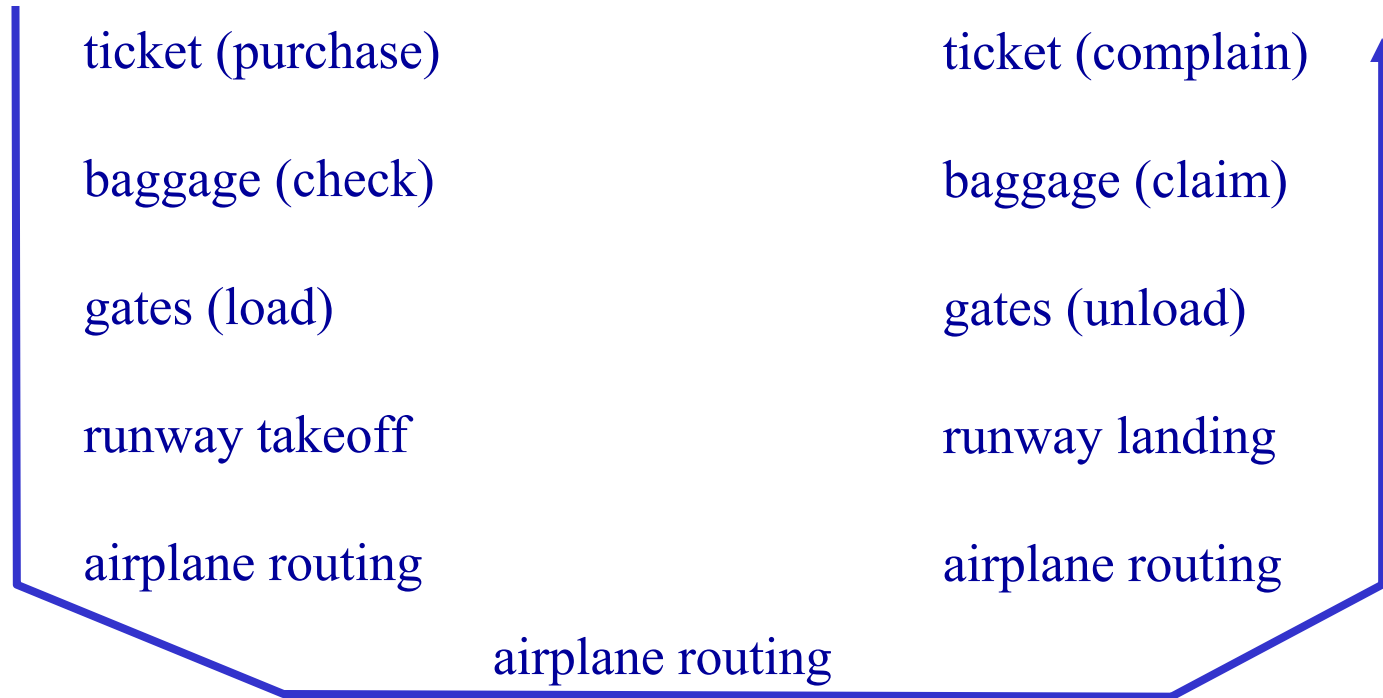
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

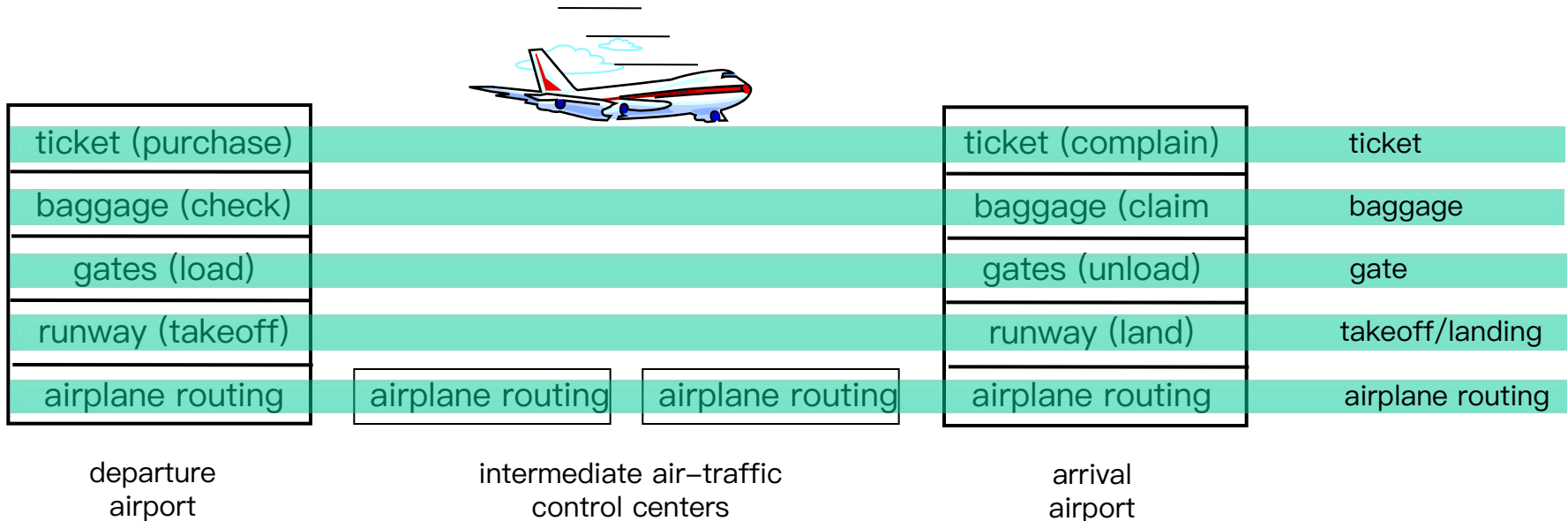
is there any hope of
organizing structure of
network?

.... or at least our
discussion of networks?

Organization of air travel



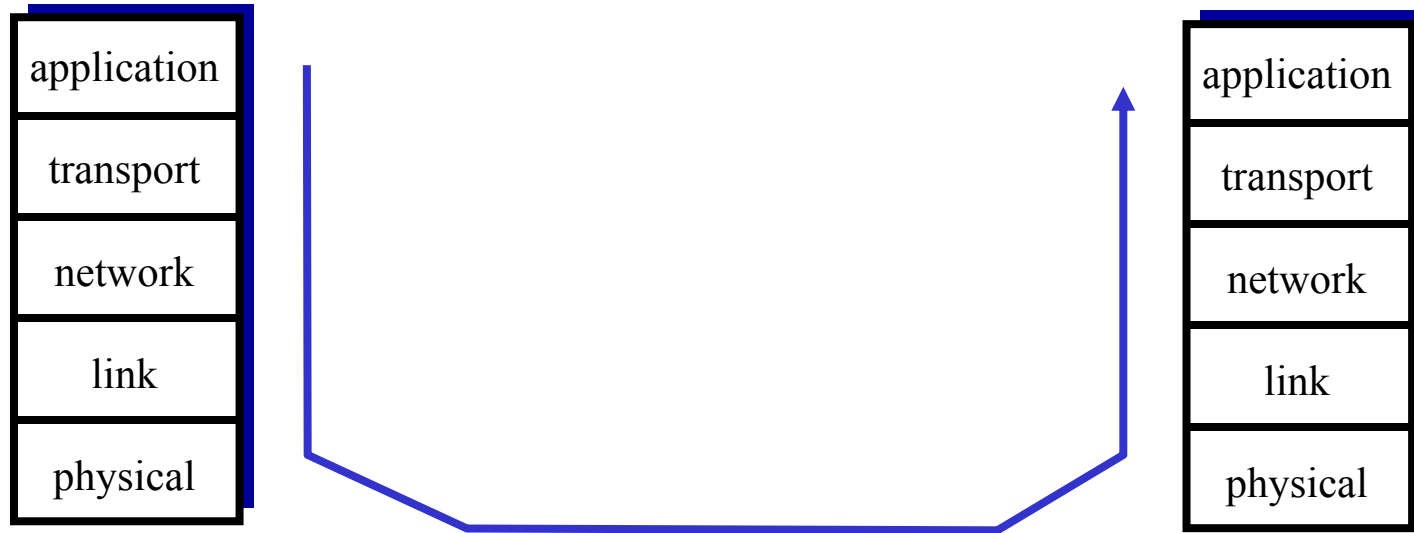
Layering of airline functionality



Layers: Each layer **provide** services **to the layer above**

- **via** its own internal-layer actions
- **relying on** services provided by layer below

Layering of functionality



Layers: Each layer **provide** services to the layer above

- **via** its own internal-layer actions
- **relying on** services provided by layer below

For example, **reliable delivery** of messages at layer n :

- Adding layer n functionality to **detect and retransmit lost messages**
- Using an **unreliable message delivery** service at layer $n-1$

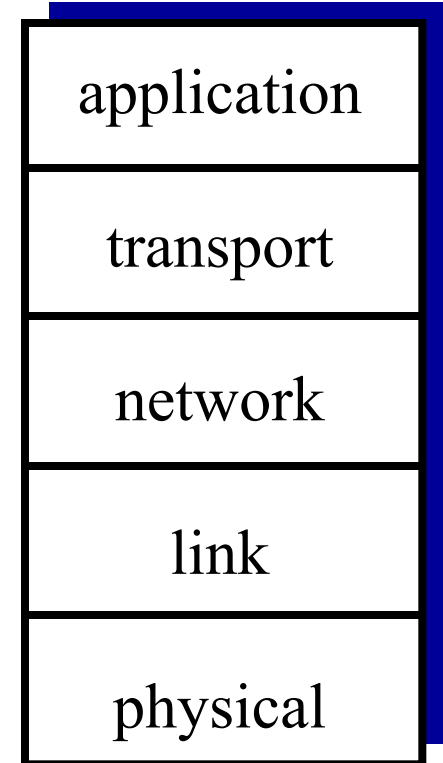
Why layering?

Dealing with complex systems:

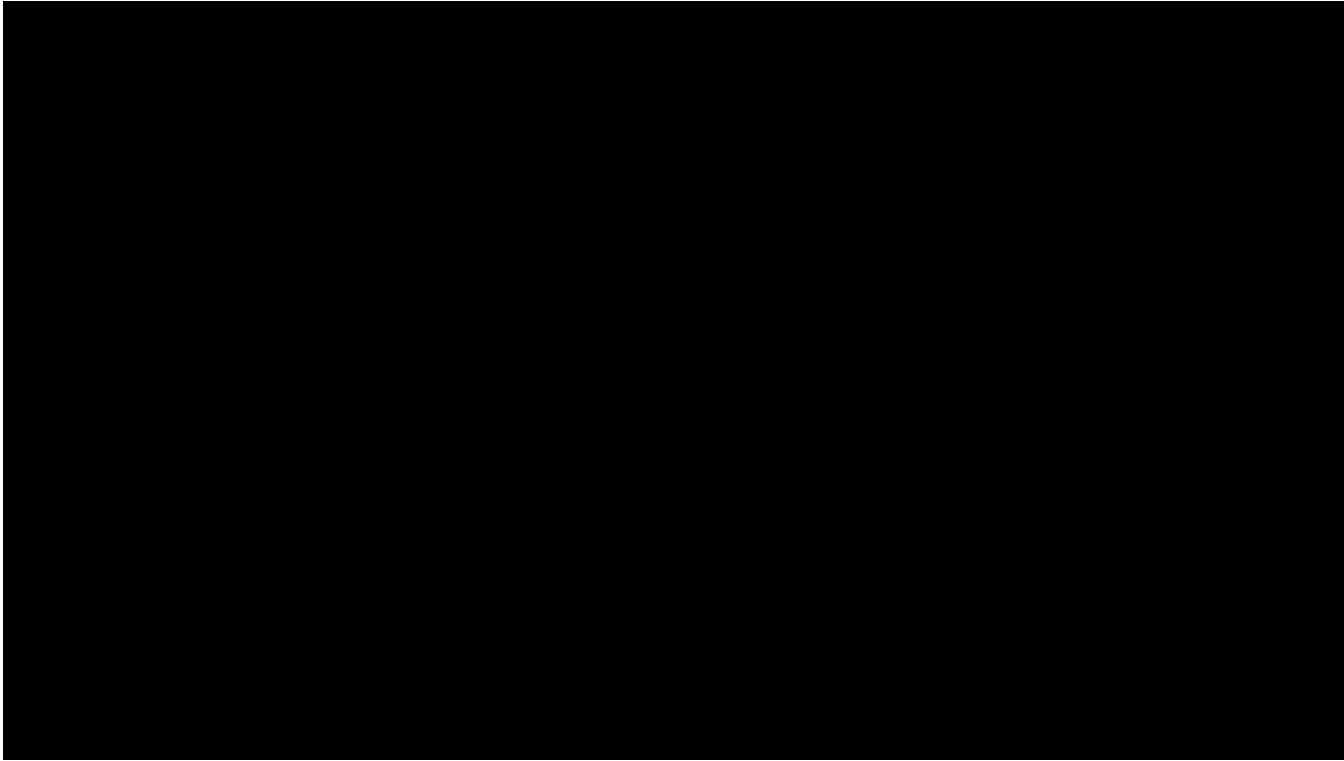
- ❖ Explicit structure allows identification, relationship of complex system's pieces
 - layered reference model for discussion
- ❖ modularization eases maintenance, updating of system
 - change of implementation of layer's service **transparent** to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- ❖ drawback?
 - One layer may duplicate lower layer functionality
 - Functionality at one layer may need information that is present only in another layer

Internet protocol stack

- ❖ **Application:** supporting network applications
 - FTP, SMTP, HTTP
- ❖ **Transport:** process-process data transfer
 - TCP, UDP
- ❖ **Network:** routing of datagrams from source to destination
 - IP, routing protocols
- ❖ **Link:** data transfer between neighboring network elements
 - Ethernet, 802.111 (WiFi), PPP
- ❖ **Physical:** bits “on the wire”

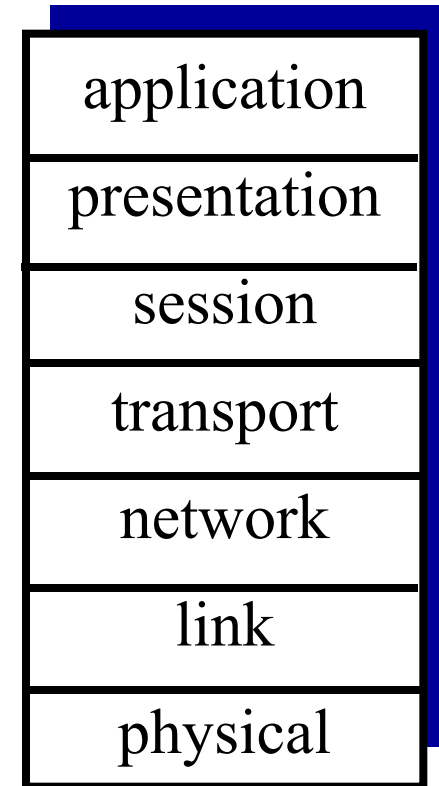


Internet protocol stack

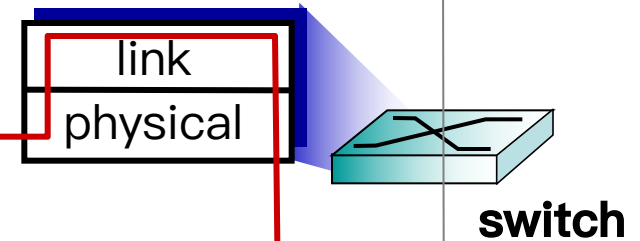
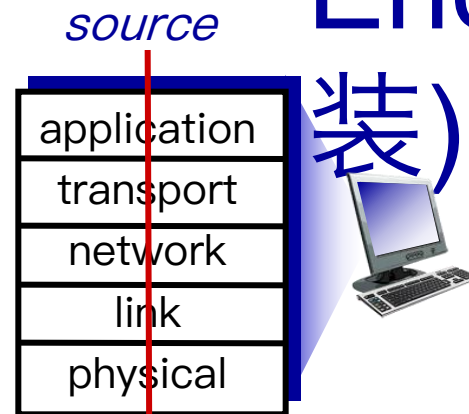
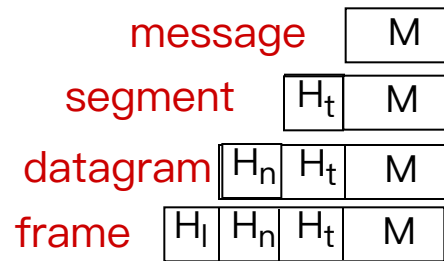


ISO/OSI reference model

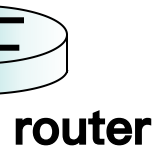
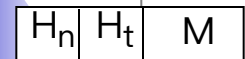
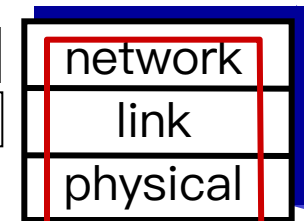
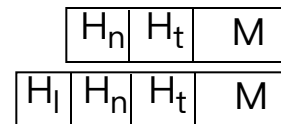
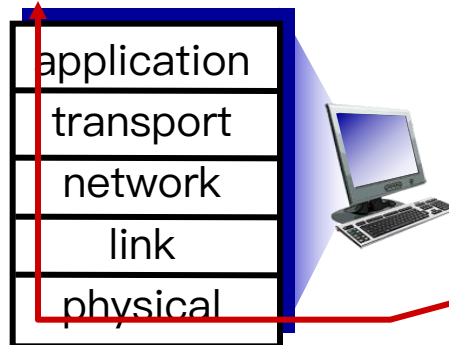
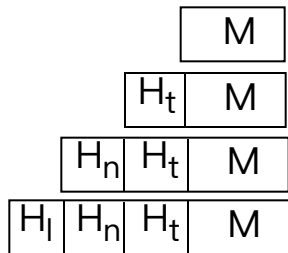
- ❖ **Presentation:** allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❖ **Session:** synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in **application**
 - needed?



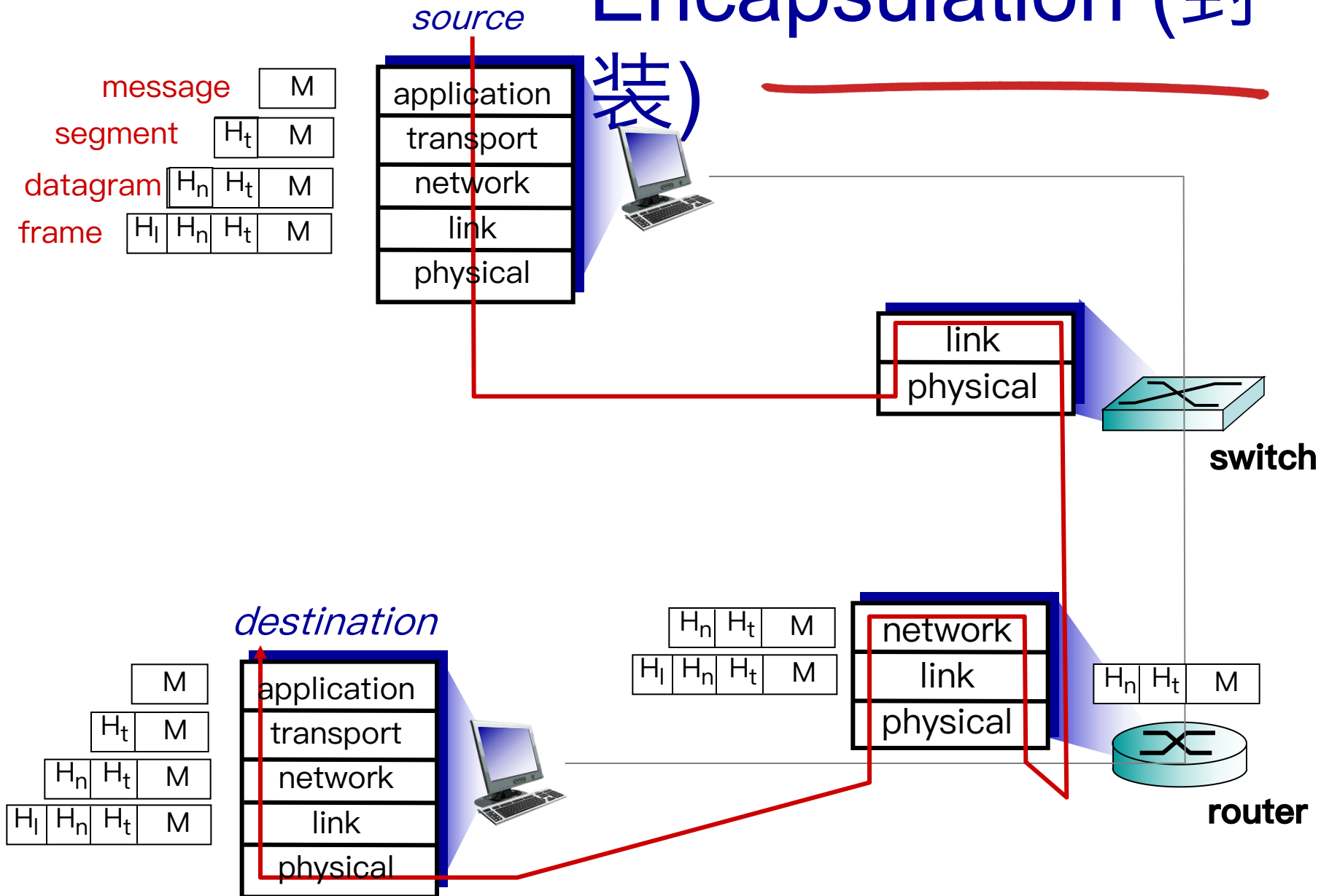
Encapsulation (封装)



destination



Encapsulation (封装)



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Network security

We attach devices to the Internet because we want to **receive/send data** from/to the Internet

- ❖ **Field of network security:**
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks
- ❖ **Internet not originally designed with (much) security in mind**
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!

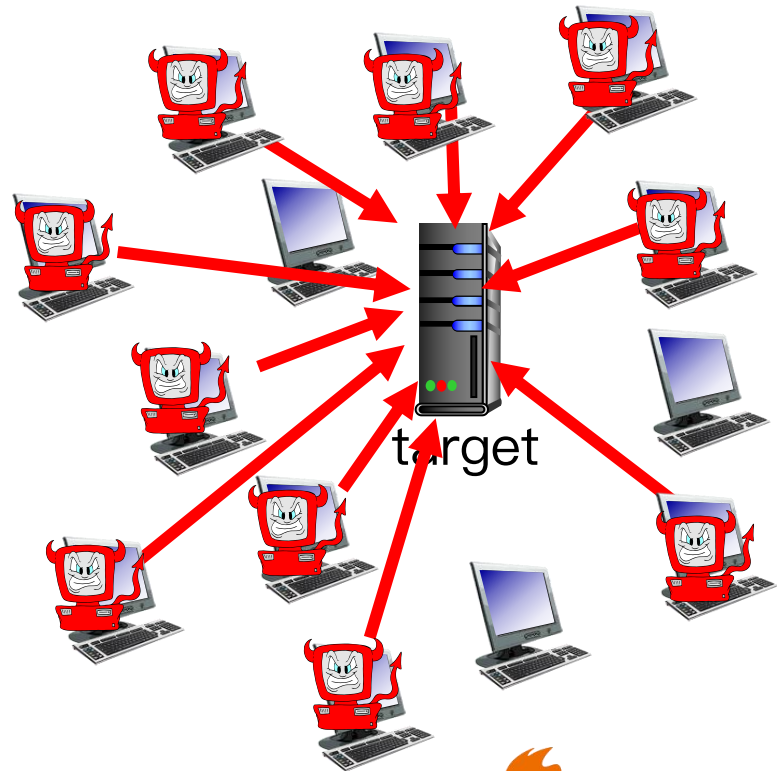
Bad guys: put malware (恶意软件) into hosts via Internet

- ❖ malware can get in host from:
 - *virus*: self-replicating infection by receiving/executing object (e.g., e-mail attachment) – **with** user interaction
 - *worm*: self-replicating infection by passively receiving object that gets itself executed – **without** user interaction
- ❖ **spyware malware** can record keystrokes, web sites visited, upload info to collection site
- ❖ infected host can be enrolled in **botnet** (僵尸网络), used for spam. DDoS attacks

Bad guys: attack server, network infrastructure

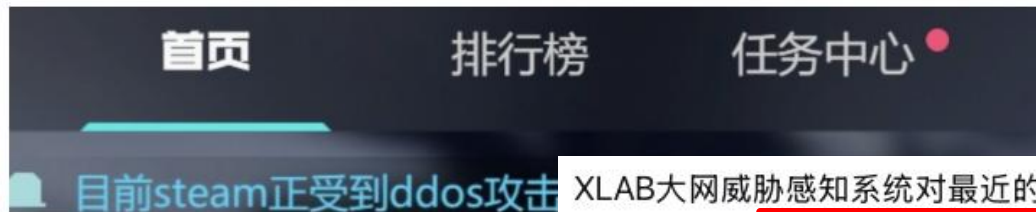
Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by **overwhelming resource** with bogus traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



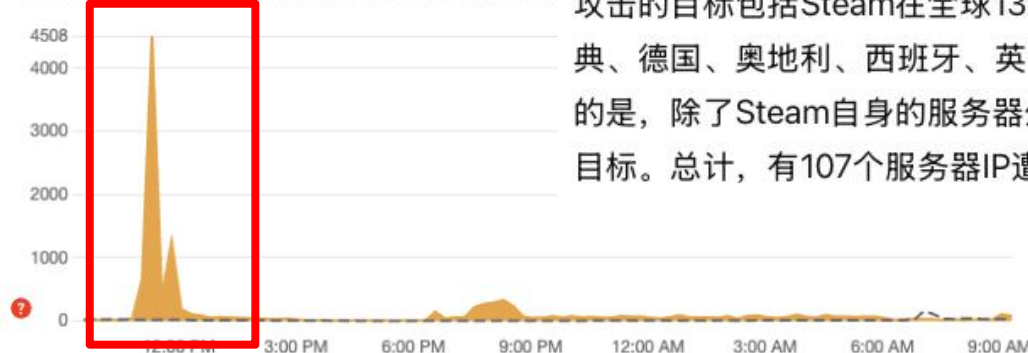
Bad guys: attack server, network infrastructure

8月24日晚，Steam平台突然崩溃，国内外玩家纷纷反馈无法登录。许多玩家猜测崩溃是由于《黑神话：悟空》在线人数过多导致。然而，根据完美世界竞技平台的公告，此次Steam崩溃实际上是因为遭受了大规模DDoS攻击。



完美世界：XLAB大网威胁感知系统对最近的DDoS攻击事件进行了深入观察。我们注意到，此次攻击涉及了近60个僵尸网络主控节点，这一规模远超过常规僵尸网络的控制范围。这些主控节点协同指挥了大量被感染的bots，以波次方式发起了攻击。

Steam outages reported in the last 24 hour

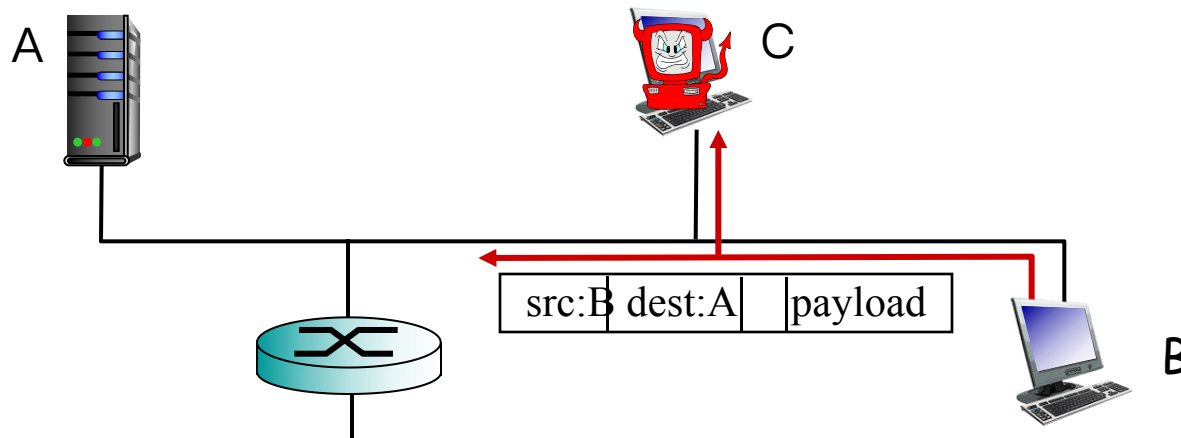


攻击的目标包括Steam在全球13个地区的服务器IP，包括中国、美国、新加坡、瑞典、德国、奥地利、西班牙、英国、日本、韩国、澳大利亚、智利和荷兰。值得注意的是，除了Steam自身的服务器外，国内完美世界代理的Steam服务器也被列为攻击目标。总计，有107个服务器IP遭到了攻击。

Bad guys can sniff packets

Packet “sniffing” :

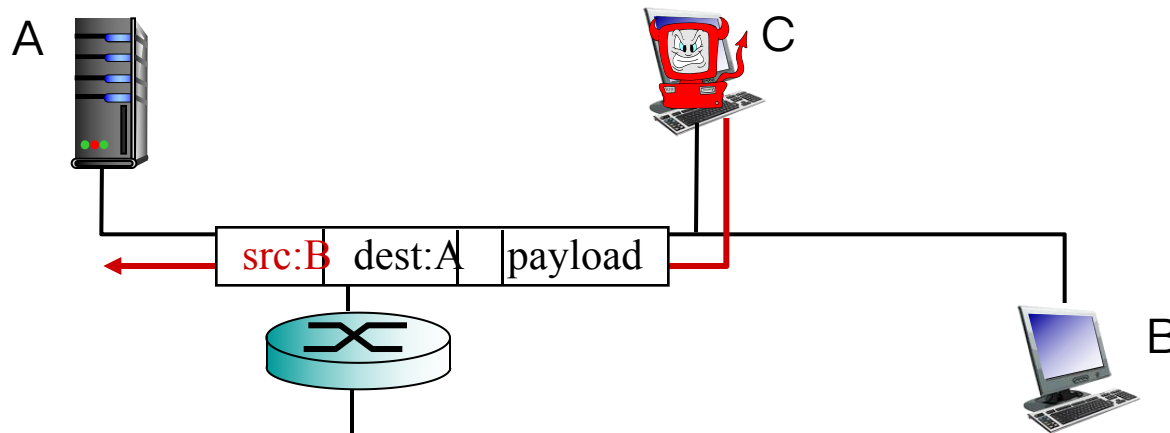
- Broadcast media (shared ethernet, wireless)
- Reads/records all packets (e.g., including passwords!) passing by
- They are difficult to detect



- ❖ wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: send packet with false source address



Network under Attack



simply learn

Lines of defense:

- **authentication:** proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **confidentiality:** via encryption
- **integrity checks:** digital signatures prevent/detect tampering
- **access restrictions:** password-protected VPNs
- **firewalls:** specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks

... lots more on security (throughout, Chapter 8)

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security